Implementation Plan

Foundational Platform Service v2.0

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1. Phased Milestones

Phase Overview

The implementation follows a three-phase approach, progressing from foundational AWS infrastructure through multi-cloud extension to full production certification.

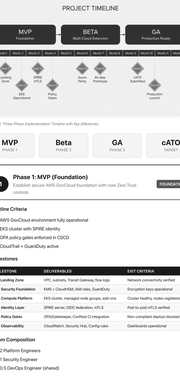

Figure 1.1: Three-Phase Implementation Timeline with Key Milestones

MVP	Beta	GA	cATO
PHASE 1	PHASE 2	PHASE 3	TARGET

Phase 1: MVP (Foundation)

FOUNDATION

Establish secure AWS GovCloud foundation with core Zero Trust controls

Outline Criteria

- AWS GovCloud environment fully operational
- EKS cluster with SPIRE identity
- GPA policy gates enforced in CI/CD
- CloudTrail + GuardDuty active

Milestones

MILESTONE	DELIVERABLES	EXIT CRITERIA
1.1 Landing Zone	VPC, subnets, Transit Gateway, flow logs	Network connectivity verified
1.2 Security Foundation	KMS + CloudHSM, IAM roles, GuardDuty	Encryption keys operational
1.3 Compute Platform	EKS cluster, managed node groups, add-ons	Cluster healthy, nodes registered
1.4 Identity Layer	SPIRE server, OIDC federation, mTLS	Pod-to-pod mTLS verified
1.5 Policy Gates	OPA/Gatekeeper, Conftest CI integration	Non-compliant deploys blocked
1.6 Observability	CloudWatch, Security Hub, Config rules	Dashboards operational

Team Composition

- 2 Platform Engineers
- 1 Security Engineer
- 0.5 DevOps Engineer (shared)

Phase 2: Beta (Multi-Cloud Extension)

MULTI-CLOUD

Extend platform to Azure Government and air-gapped K3s prototype

Outline Criteria

- Azure Government environment deploys from same IaC
- K3s cluster runs Zarf-packaged platform components
- SPIRE federation works across clouds
- 70% FedRAMP High controls automated

Milestones

MILESTONE	DELIVERABLES	EXIT CRITERIA
2.1 Azure Foundation	VNet, AKS, Key Vault, Defender	Azure environment operational
2.2 Multi-Cloud Identity	SPIRE federation JWT/x.509	Cross-cloud mTLS verified
2.3 Air-Gap Prototype	K3s + Zarf packages, offline registry	Disconnected deployment works
2.4 Centralized Logging	Security Lake aggregation, cross-cloud	Single pane of glass
2.5 Compliance Automation	Evidence generation, control mapping	70% controls automated
2.6 Tenant Onboarding	Self-service namespace provisioning	Tenant deployed <15 min

Team Composition

- 3 Platform Engineers
- 1 Security Engineer
- 1 DevOps Engineer
- 0.5 Compliance Analyst (shared)

Phase 3: GA (Production Ready)

PRODUCTION

Production certification with continuous ATO capability

Outline Criteria

- cATO certification achieved
- 3+ tenants onboarded successfully
- DDIL operations validated for 30 days
- DR failover tested

Milestones

MILESTONE	DELIVERABLES	EXIT CRITERIA
3.1 cATO Package	SSP, control evidence, POA&M	ATO granted
3.2 Production Hardening	Break-glass procedures, DR runbooks	Incident response tested
3.3 DDIL Validation	30-day disconnected operation test	Store-forward sync works
3.4 Tenant Migration	Production tenant onboarding	3+ tenants operational
3.5 Operational Handoff	Runbooks, training, SOP documentation	Ops team certified

Team Composition

- 2 Platform Engineers
- 1 Security Engineer
- 1 DevOps Engineer
- 1 Compliance Analyst
- 0.5 Technical Writer (shared)

2. Resourcing Plan

Roles and Responsibilities

ROLE	RESPONSIBILITIES	FTE
Platform Engineer	Terraform modules, EKS/AKS configuration, troubleshooting	2-3
Security Engineer	SPIRE deployment, threat modeling, incident response	1
DevOps Engineer	CI/CD pipelines, GitOps, monitoring	1
Compliance Analyst	Control mapping, evidence collection, ATO documentation	0.5-1
Technical Writer	PKI, architecture docs, runbooks	0.5
Platform Lead	Technical direction, stakeholder management	1

Staffing by Phase

Figure 2.1: Team Staffing Levels Across Implementation Phases

ROLE	PHASE 1 (MVP)	PHASE 2 (BETA)	PHASE 3 (GA)
Platform Engineers	2	3	2
Security Engineers	1	1	1
DevOps Engineers	0.5	1	1
Compliance Analysts	-	0.5	1
Technical Writers	-	-	0.5
FTE Total	**3.5**	**6.0**	**6.5**

3. Build vs. Buy Analysis

Component Decisions

COMPONENT	DECISION	RATIONALE	COST IMPACT
Kubernetes	Buy (EKS/AKS)	Managed control plane reduces ops burden; FedRAMP authorized	+$150/cluster/month
Service Mesh	Build (Istio)	Open source, full control, SPIRE integration	Engineering time only
Policy Engine	Build (OPA)	CNCF standard; Rego flexibility; no licensing	Engineering time only
Secrets Manager	Buy (AWS/Azure native)	FedRAMP authorized; native integration	Usage-based
SIEM	Buy (Splunk/Security Lake)	Mature product; compliance reporting	$10k-100k/year
Identity (SPIFFE)	Build (SPIRE)	Multi-cloud portable; no vendor lock-in	Engineering time only
GitOps	Build (Flux CD)	CNCF graduated; multi-tenancy model	Engineering time only
Container Registry	Buy (ECR/ACR)	Managed; vulnerability scanning; PrivateLink	Usage-based
Vulnerability Scanning	Buy (Inspector/Defender)	Native integration; continuous scanning	Included in pricing
HSM	Buy (CloudHSM)	FIPS 140-3 L3, required for IL5	$1.50/hour/HSM

Build vs. Buy Summary

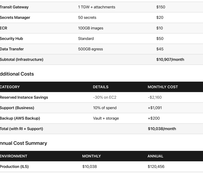

Figure 3.1: Build vs. Buy Breakdown by Category

CATEGORY	BUILD	BUY
Infrastructure	20%	80%
Security Tooling	40%	60%
Identity	70%	30%
Observability	30%	70%
Compliance	50%	50%

4. Cost Model

Notional Environment: Production IL5

Environment Assumptions

- AWS GovCloud (us-gov-west-1)
- 3 Availability Zones
- EKS with 10 worker nodes (m5.2xlarge)
- Dedicated tenancy (IL5 requirement)
- CloudHSM cluster (2 HSMs for HA)
- 1TB log retention (365 days)

Monthly Infrastructure Cost

RESOURCE	CONFIGURATION	MONTHLY COST
EKS Cluster	1 cluster	$73
EC2 Worker Nodes	10x m5.2xlarge dedicated	$7,200
NAT Gateways	3 (HA across AZs)	$324
VPC Endpoints	15 endpoints	$225
Application Load Balancer	1 ALB	$100
CloudHSM	2 HSMs (HA)	$2,160
KMS	5 CMKs + usage	$50
S3 (Logs)	1TB + requests	$150
CloudTrail	Management + data events	$200
GuardDuty	Standard + K8s	$100
Config	Rules + evaluations	$50
Transit Gateway	1 TGW + attachments	$150
Secrets Manager	50 secrets	$20
ECR	10GB images	$10
Security Hub	Standard	$50
Data Transfer	500GB egress	$45
Subtotal (Infrastructure)		**$10,907/month**

Additional Costs

CATEGORY	DETAILS	MONTHLY COST
Reserved Instance Savings	-30% on EC2	-$2,160
Support (Business)	10% of spend	+$1,091
Backup (AWS Backup)	Vault + storage	+$200
Total (with RI + Support)		**$10,038/month**

Annual Cost Summary

ENVIRONMENT	MONTHLY	ANNUAL
Production (IL5)	$10,038	$120,456
Staging	$5,019	$60,228
Development	$3,346	$40,152
Air-Gap Edge	$0 (hardware CapEx)	$50,000 (one-time)
Total Annual Platform Cost		**~$271,000**

Cost Optimization Opportunities

OPTIMIZATION	POTENTIAL SAVINGS
Spot instances for dev/staging	60-70% on compute
Reserved instances (1-year)	30-40% on production
S3 Intelligent Tiering	20-30% on log storage
Right-sizing after baseline	10-20% on compute

5. Timeline Summary

The implementation spans 12 months with clear phase boundaries and milestone checkpoints throughout.

MILESTONE	WEEK	PHASE	KEY DELIVERABLE
M1: Landing Zone	Week 2	MVP	VPC/Network infrastructure operational
M2: EKS Operational	Week 6	MVP	Kubernetes cluster healthy and accessible
M3: SPIRE + mTLS	Week 10	MVP	Workload identity and encryption active
M4: Policy Gates Live	Week 14	MVP	OPA admission control enforcing policies
M5: Azure Parity	Week 22	Beta	Azure Government environment operational
M6: Air-Gap Prototype	Week 26	Beta	K3s disconnected deployment working
M7: cATO Submitted	Week 34	GA	Complete ATO package submitted for review
M8: Production Launch	Week 40	GA	Platform certified and tenants onboarded

6. Risk Mitigation Schedule

RISK	MITIGATION	OWNER	DUE

HSM provisioning delay	Order CloudHSM in Week 1	Security	Week 1
SPIRE complexity	PoC in Week 2-3 before full deployment	Platform	Week 3
Azure subscription delays	Escalate to leadership if not resolved by Week 12	Platform Lead	Week 12
CDS API spec delays	Design abstract interface; defer CDS integration	Platform	Week 16
ATO documentation gaps	Engage compliance team early; weekly syncs	Compliance	Ongoing

7. Stakeholder Communication Plan

STAKEHOLDER	CADENCE	FORMAT	OWNER
Executive Sponsor	Monthly	Status report + demo	Platform Lead
Security/ISSO	Bi-weekly	Compliance sync	Security Engineer
Tenant Teams	Weekly	Office hours	Platform Engineer
Operations	Daily (post-Beta)	Standup + handoff	DevOps
AO/Accreditor	As needed	ATO package reviews	Compliance Analyst